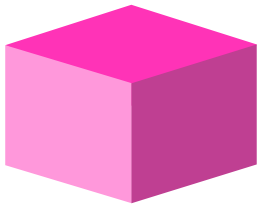
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Topic 2. 2:
Energy Sources

Form of Energy Sources

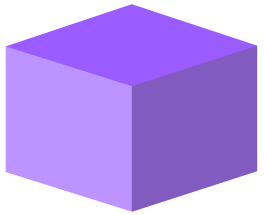


01

NON-RENEWABLE

deplete

- Energy sources that cannot be replaced and it reserves eventually run out
- Much cheaper than renewable sources but are harmful to the environment



02

RENEWABLE

- Energy sources that can be replaced quickly and will not deplete
- Usually more expensive to obtain than renewable sources but are not harmful to the environment

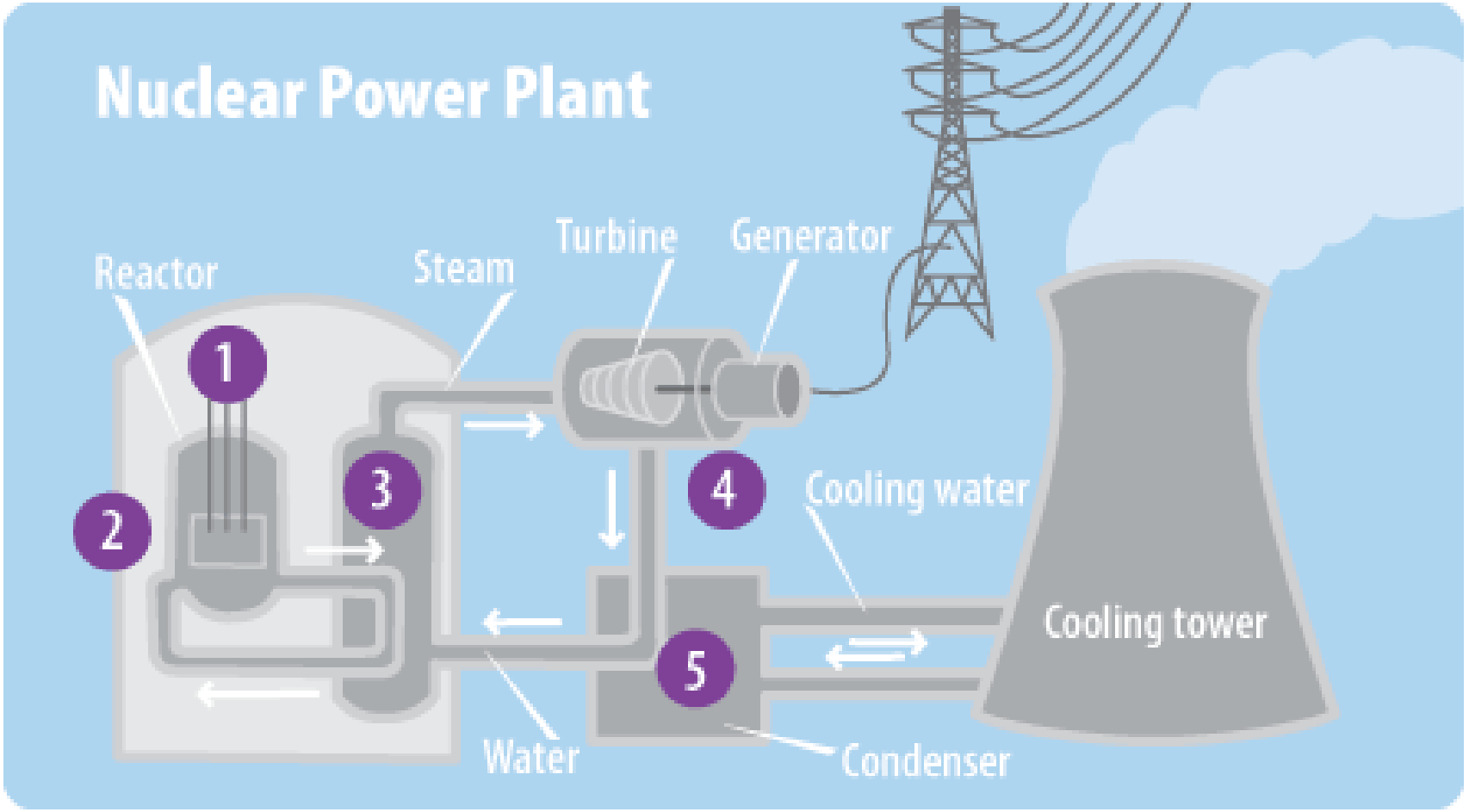
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NON-RENEWABLE ENERGY

Nuclear Energy

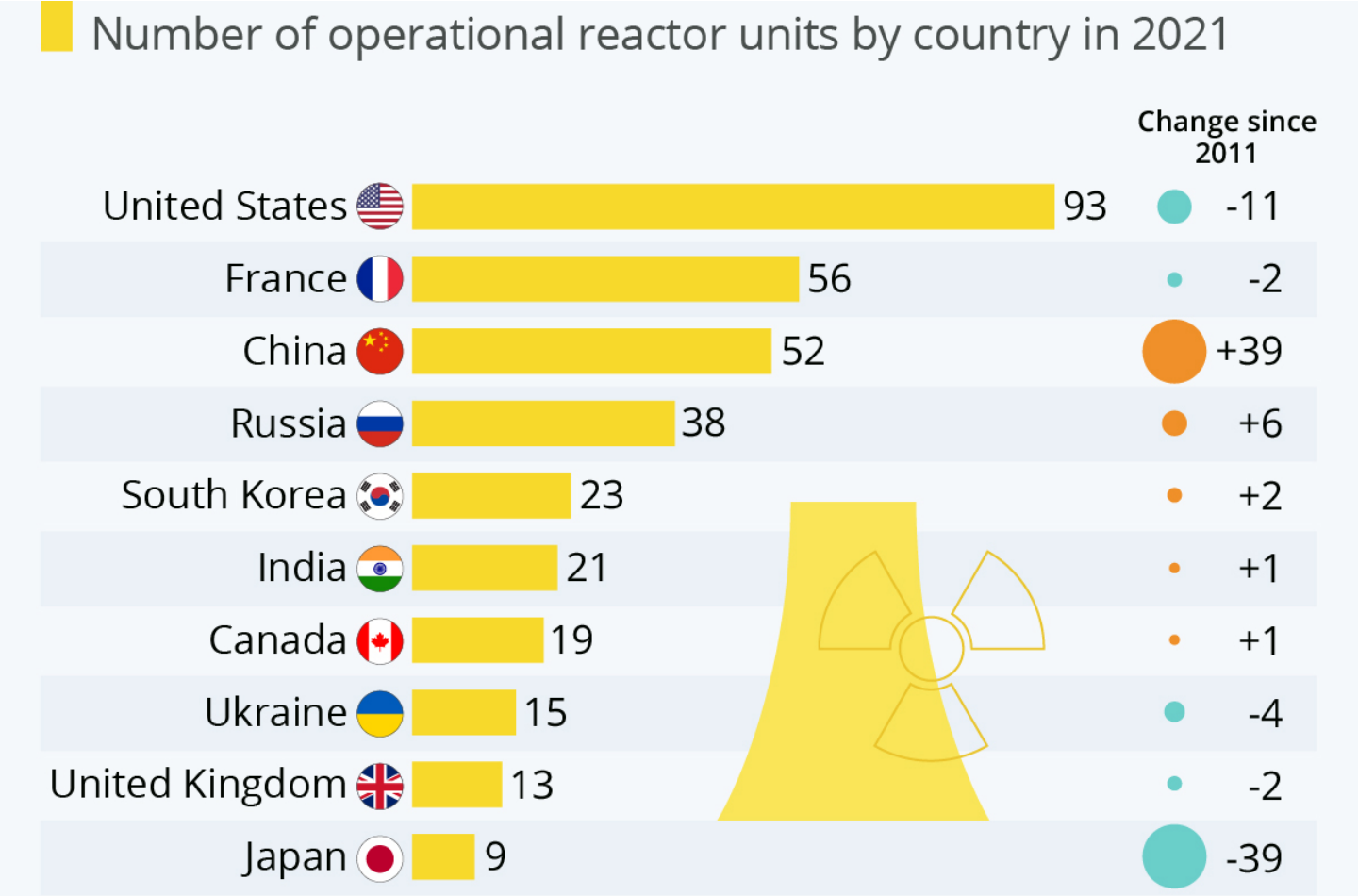
- The source which generate nuclear energy comes from **uranium**, which are found throughout the earth's crust, but most of it is too difficult or too expensive to mine and process into fuel for nuclear power plants.
- In all power plants, water is heated to produce steam, which turns a generator to create electricity. Similarly, in nuclear power stations, instead of using coal or diesel to heat the water to produce steam, steam in nuclear power plant is made by the heat generated from a type of nuclear reaction called **nuclear fission**.

Nuclear Energy



Source: https://archive.epa.gov/climatechange/kids/solutions/technologies/nuclear.html

Top 10 Nuclear Energy Producers



Why Nuclear Energy?

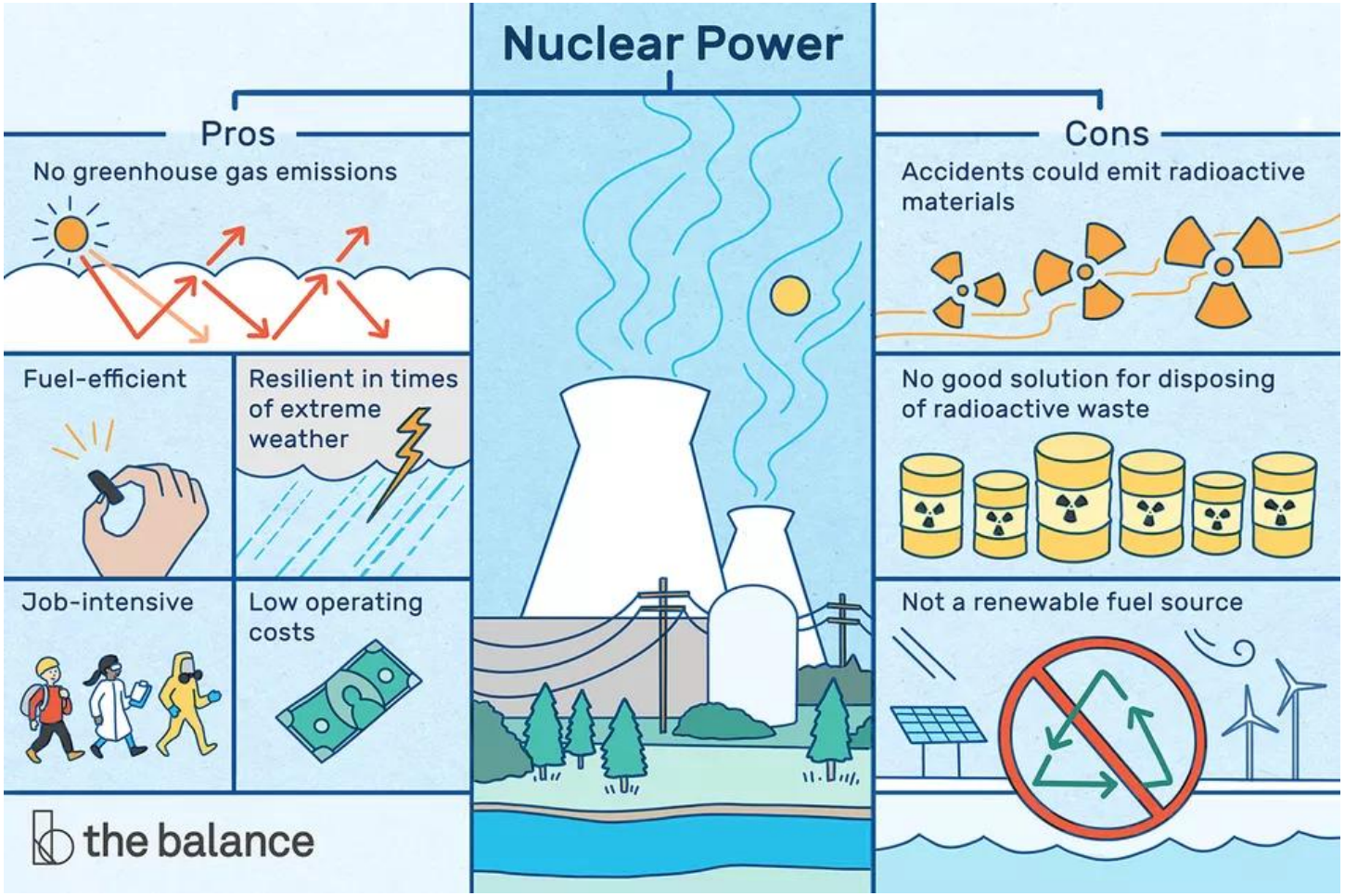
- In comparison to a traditional fossil fuel such as coal, nuclear fuel generates millions of times more power.
- For example 1kg of coal would power a 60 watt light bulb for four days. In comparison, nuclear would power the same bulb for 685 years.
- Nuclear also saves billions of tones of CO_2 emissions. Compared to coal, the power generated by existing nuclear power stations avoids 40 million tones of CO_2 per year

Nuclear Energy

Nuclear Power Plant in Malaysia

https://www.nst.com.my/news/government-public-policy/2020/02/564295/malaysia-wont-use-nuclear-power-says-pm

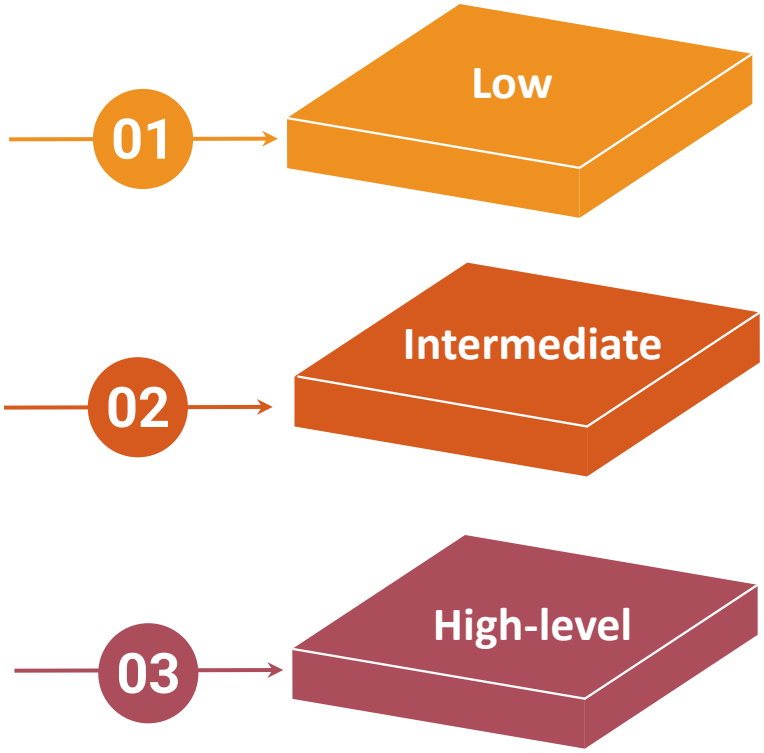
Nuclear Energy

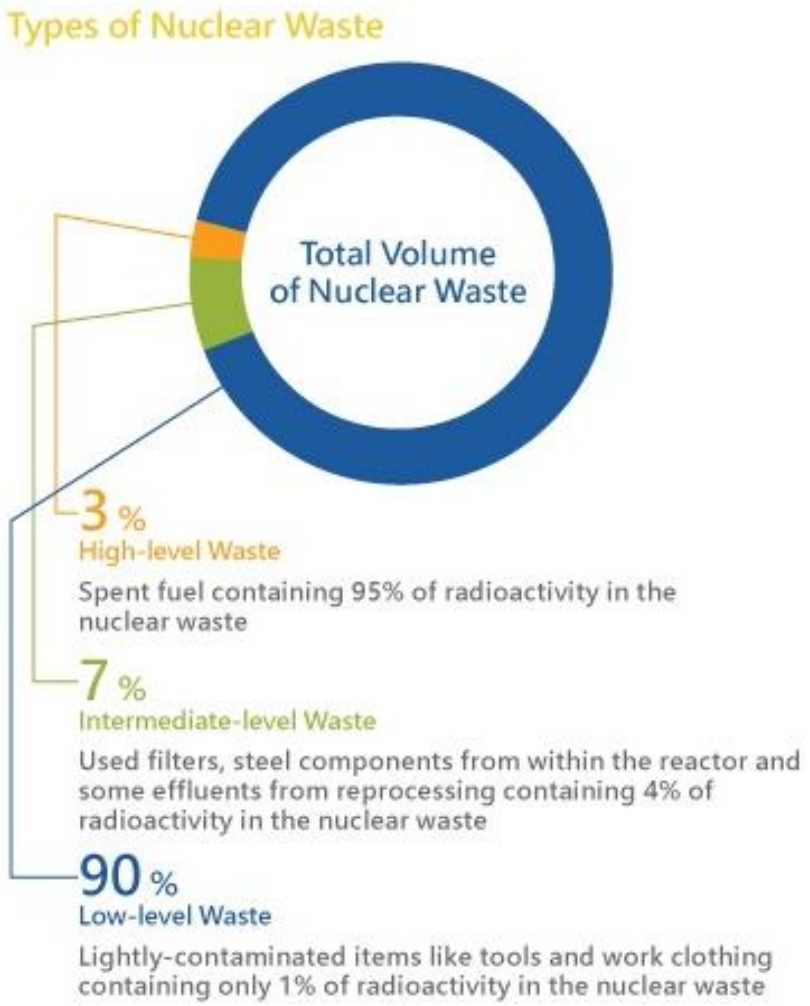


Source: https://www.thebalance.com/nuclear-power-how-it-works-pros-cons-impact-3306336

Waste from Nuclear Power Plant

They are classified according to their radioactivity level



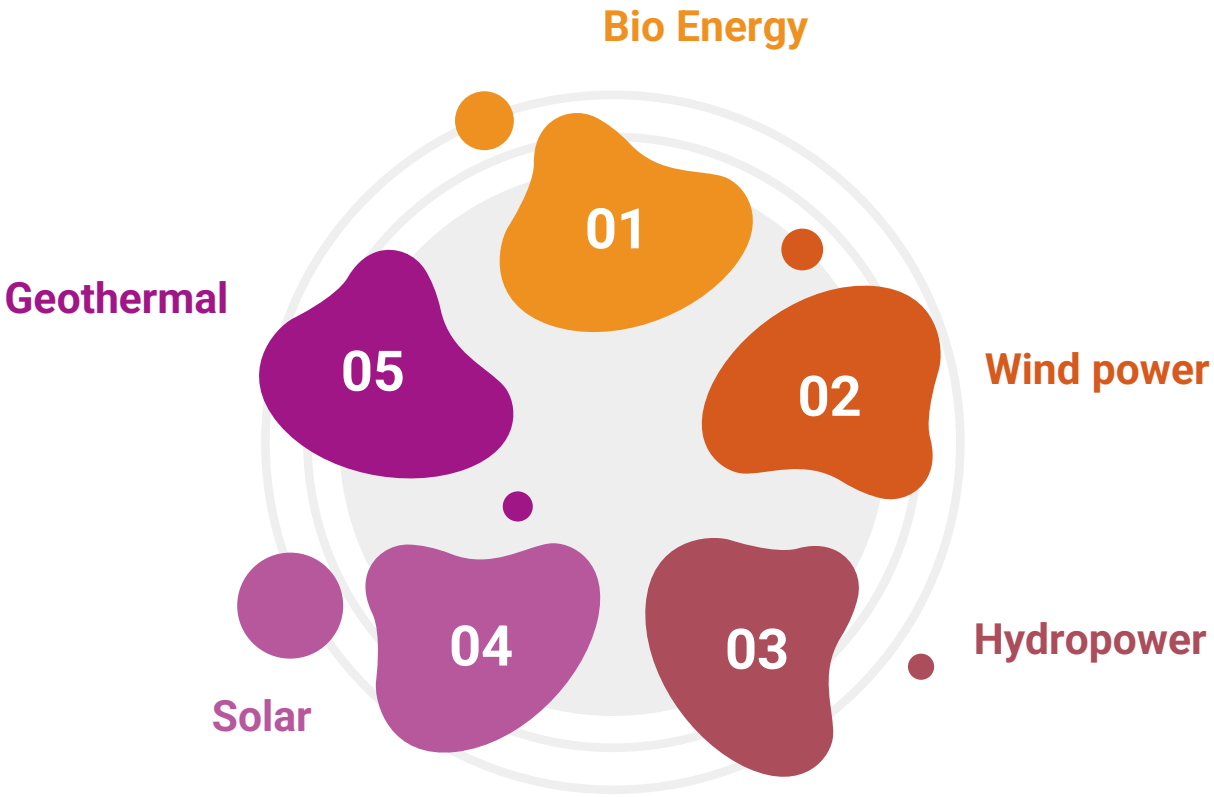


Source: https://world-nuclear.org/nuclear-essentials/what-is-nuclear-waste-and-what-do-we-do-with-it.aspx

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RENEWABLE ENERGY

Renewable Energy Sources



Bio Energy

- Consists of **three (3)** main types:

1. Biomass

- Biomass is biological material derived from living, or recently living organisms.
- Wood remains the largest biomass energy source today. Examples include forest entities – such as dead trees, branches and tree stumps
- Plant energy is produced by crops specifically grown for use as fuel that offer high biomass output per hectare with low input energy

2. Biogas

- Rotting garbage, and agricultural and human waste, all release methane gas – also called landfill gas or biogas
- Biomass can be converted to other usable forms of energy like methane gas or transportation fuels like ethanol and biodiesel.
- Crops, such as corn and sugarcane, can be fermented to produce the transportation fuel, ethanol.

3. Biofuel

- Biofuels include a wide range of fuels which are derived from biomass. The term covers solid, liquid, and gaseous fuels.
- Liquid biofuels include bio alcohols, such as bioethanol, and bio oils, such as biodiesel.
- Biodiesel, another transportation fuel, can be produced from left-over food products like vegetable oils and animal fats
- Bioethanol is an alcohol made by fermenting the sugar components of plant materials and it is made mostly from sugar and starch crops

Solar Energy

- Types of solar energy

1. Passive

- Depends heavily on the design, construction, and building of a structure.
- Uses the sun's energy for heating and cooling purposes without reliance on external devices.
- Uses passive collectors to convert rays into the sunlight, which are basically based on laws of thermodynamics which transfer heat from warmer to cooler surfaces.

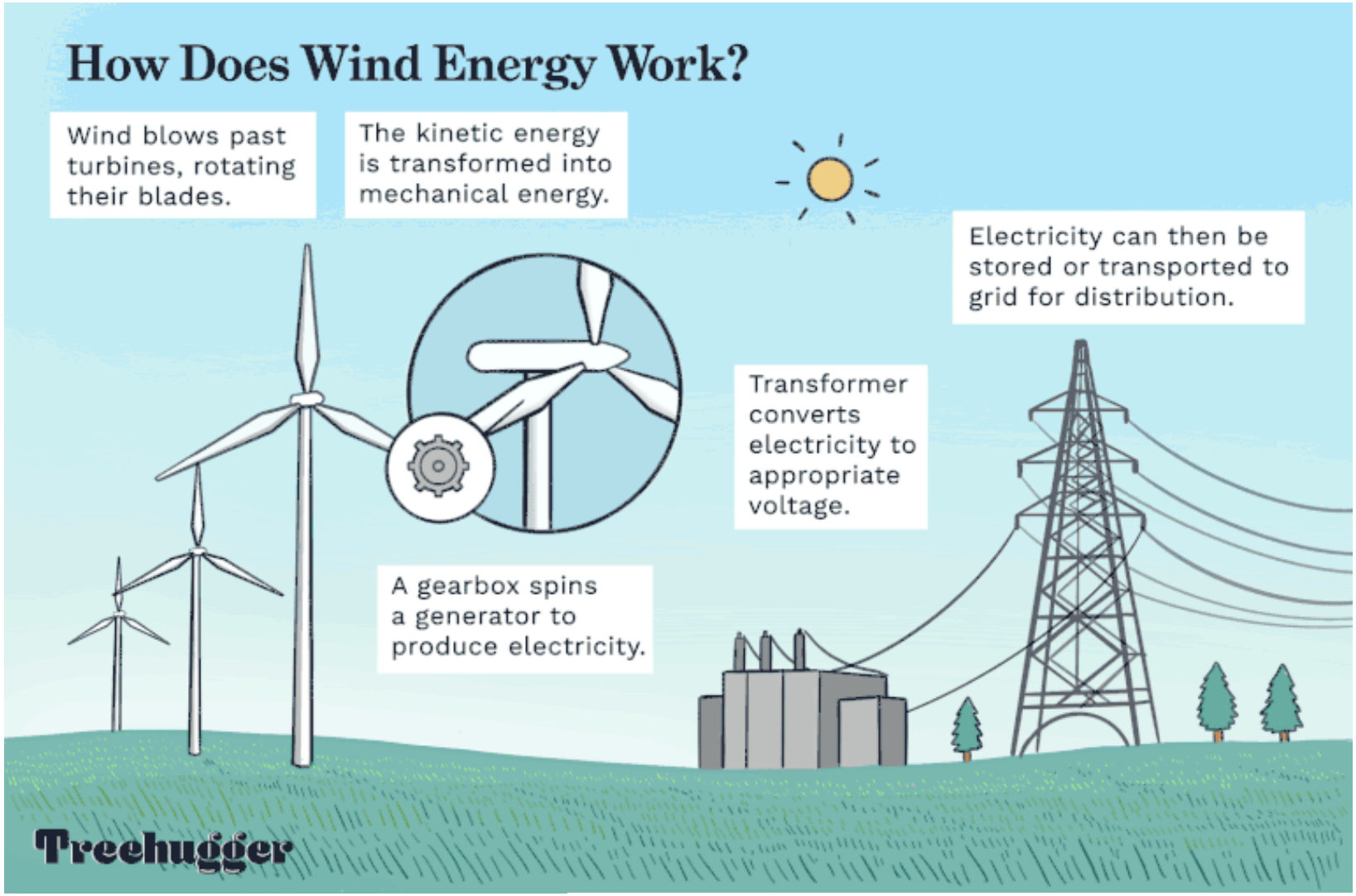
2. Active

- Active systems have devices to convert the sun's energy into a more usable form, such as hot water or electricity.
- The way active solar energy works is by using air or liquid to capture the sun's heat. The fluid is then moved to a storage container until it is converted to energy. Active solar energy systems must use external devices to collect, store, and convert the sun's solar energy into usable power (i.e. solar panels).
- Liquid is commonly used because it is a better conductor of heat and energy. However, air has the benefit that it doesn't freeze. Both fluids can be used in the heating and cooling of a home or building. Liquid collectors are called hydronic collectors while air systems are simply called air collectors.

Wind Power

- Wind energy is electricity created from the naturally flowing air in the Earth's atmosphere using large turbines to take available wind.
- Areas where winds are stronger and more constant, such as offshore and high altitude sites, are preferred locations for wind farms.
- Turbines with rated output of **1.5 to 3 MW** have become the most common for commercial use

Wind Power



Source:
https://www.treehugger.com/what-is-wind-energy-5097217

Wind Power

- Consists of **three (3)** main types:

1. Utility Scale



- These are large-scale wind projects designed to be used as a source of energy for a utility company.
- They are similar in scope to a coal-fired or natural gas power plant, which they sometimes replace or supplement.
- Turbines exceed 100 kilowatts of power in size and are usually installed in groups to provide significant power

2. Offshore



- These are generally utility-scale wind energy projects that are planned in the waters off coastal areas.
- Wind blows more consistently and strongly in offshore areas than in land
- It is estimated that the potential for offshore wind energy in the U.S. is more than 2,000 gigawatts of power, which is two times the generating capacity of all U.S. electric power plants.

3. Distributed

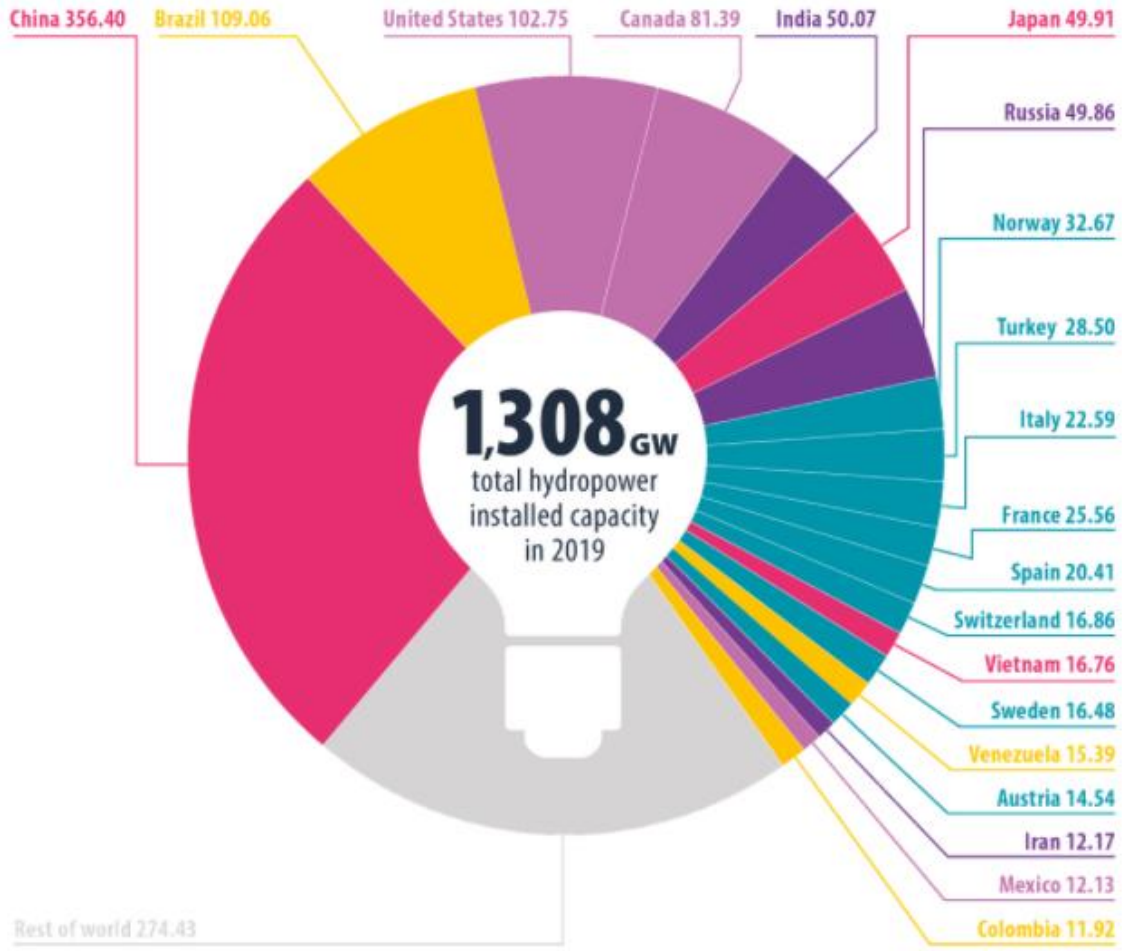


- Smaller in physical size and are used to meet the energy demands of a specific site or local area.
- Sometimes, these turbines are connected to the larger energy distribution grid, and sometimes they are off-grid.
- Smaller installations (5 kilowatt size) in residential settings, where they might provide some or most of a home's needs and medium-sized versions (20 kilowatts or so) at industrial or community sites, where they might be part of a renewable energy system

Source:
https://www.treehugger.com/what-is-wind-energy-5097217

Hydropower

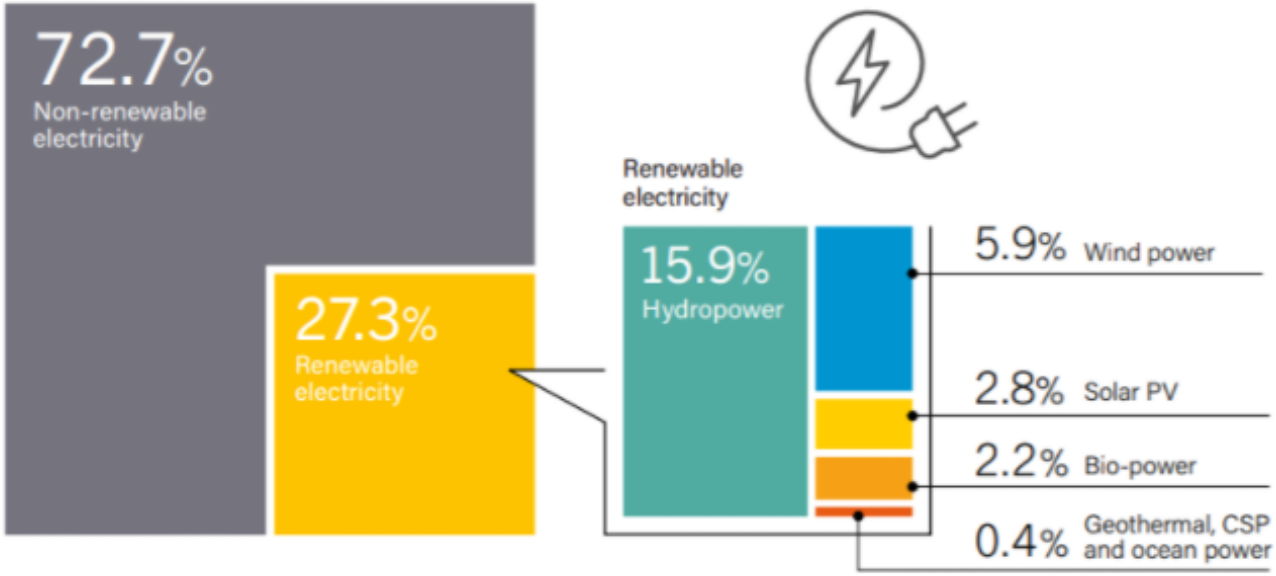
- Hydropower installed capacity reached **1,308 gigawatts (GW)** in 2019 as generation hit a record **4,306 terawatt hours** (TWh), with the average year-on-year worldwide growth in hydropower installed capacity was 2.1 per cent between 2015 and 2019.



Source: https://www.hydropower.org/iha/discover-facts-about-hydropower

Hydropower

- Hydropower produces about **16 per cent** of total electricity generation from all sources (renewable and non-renewable) in 2019.
- Around **60 per cent** of all renewable electricity is generated by hydropower.



Estimated renewable energy share of global electricity production in 2019 (Source: REN21 2020)

Source: https://www.hydropower.org/iha/discover-facts-about-hydropower

Hydropower

- Consists of **four (4)** main types:

1. Run-of-river

- Facility that channels flowing water from a river through a canal or penstock to spin a turbine.
- Provides a continuous supply of electricity (base load), with some flexibility of operation for daily fluctuations in demand through water flow that is regulated by the facility

2. Storage

- Typically a large system that uses a dam to store water in a reservoir.
- Electricity is produced by releasing water from the reservoir through a turbine, which activates a generator.

3. Pumped storage

- Provides peak-load supply, harnessing water which is cycled between a lower and upper reservoir by pumps which use surplus energy from the system at times of low demand.
- When electricity demand is high, water is released back to the lower reservoir through turbines to produce electricity

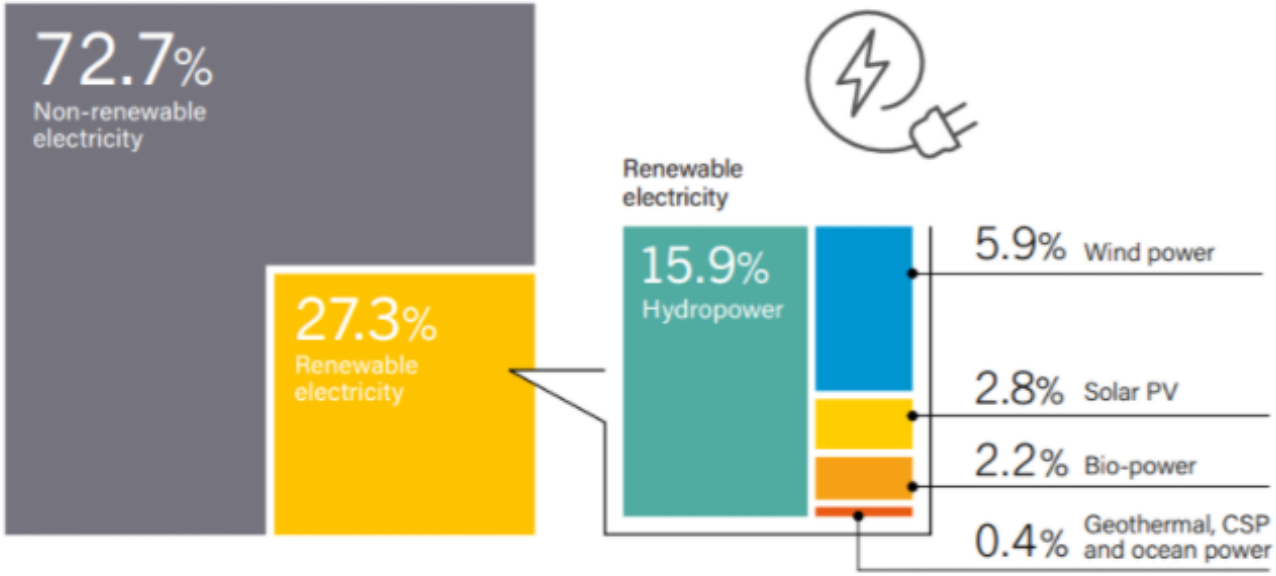
4. Offshore

- A less established but growing group of technologies that use tidal currents or the power of waves to generate electricity from seawater

Source:
https://www.hydropower.org/iha/discover-types-of-hydropower

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Geothermal Energy

- High Temperature Geothermal energy is from thermal energy generated and stored in the Earth core.
- Thermal energy is the energy that determines the temperature of matter. Earth's geothermal energy originates from the original formation of the planet and from radioactive decay of minerals
- From hot springs, geothermal energy has been used for bathing and for space heating since ancient Roman times, but it is now better known for electricity generation.



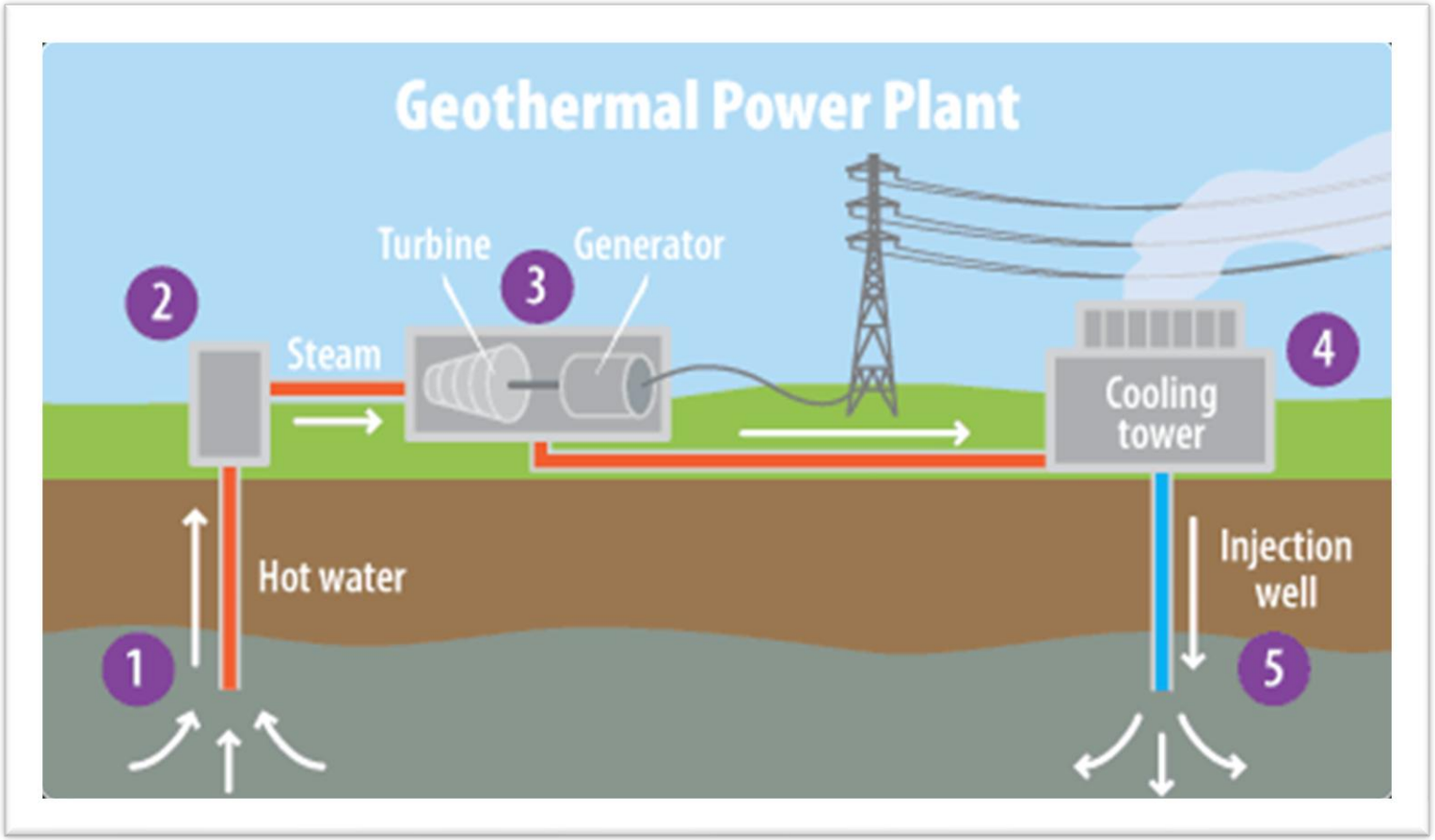




Geothermal Energy

- Geothermal energy is the energy that is produced from **beneath the earth**.
- High temperatures are produced continuously inside the earth's crust by the slow delay of radioactive particles. Hot rocks present below the earth heats up the water that produces steam. The steam is then captured, which helps to move turbines. The rotating turbines then power the generators.
- Geothermal energy can be used by a residential unit or on a large scale by an industrial application.
- Geothermal plants typically have low emissions if they pump the steam and water they use back into the reservoir.

Geothermal Energy



Source:
https://archive.epa.gov/climatechange/kids/solutions/technologies/geothermal.html



THANK you